

RBS Project - R

Telco Customer Churn

Project Questions and Answers

The eight project questions with answers, explanations, key statistics and visuals

Data: IBM Telco Customer Churn dataset (7,043 customers, California). 26.5% of customers churned (1,869 customers). All numbers and plots in this document come from the R Markdown file *Telco_Churn_EDA.Rmd*.

Part 2 - Project Questions and Answers

The assignment asked for an independent exploratory analysis of the Telco churn data, with our own analytical questions, appropriate statistics, visualizations, and a summary of findings. Below are the eight questions we chose, each with a short answer, the explanation and the key visuals.

Overview of the statistical evidence

Question	Test	Result
Q2 services	Trend test (prop.trend.test)	X-squared = 461.2, $p < 2.2e-16$
Q2 services	Chi-square, 0-1 vs 2+ services	X-squared = 301.4, $p < 2.2e-16$
Q3 payment	Chi-square payment method	X-squared = 648.1, $p < 2.2e-16$
Q3 billing	Chi-square paperless billing	X-squared = 258.3, $p < 2.2e-16$
Q4 senior	Chi-square	X-squared = 159.4, $p < 2.2e-16$
Q4 partner	Chi-square	X-squared = 158.7, $p < 2.2e-16$
Q4 dependents	Chi-square	X-squared = 433.7, $p < 2.2e-16$
Q4 gender	Chi-square	X-squared = 0.48, $p = 0.49$ (not significant)
Q5 DSL / fiber optic	Wilcoxon, charges by churn	$p < 2.2e-16$ (churners pay less)
Q5 no internet	Wilcoxon, charges by churn	$p = 0.004$ (difference only \$0.70)
Q6 cities	Chi-square, 8 cities	X-squared = 4.43, $p = 0.73$ (not significant)
Q7 CLTV	Wilcoxon, CLTV by churn	$p < 2.2e-16$

Question 1

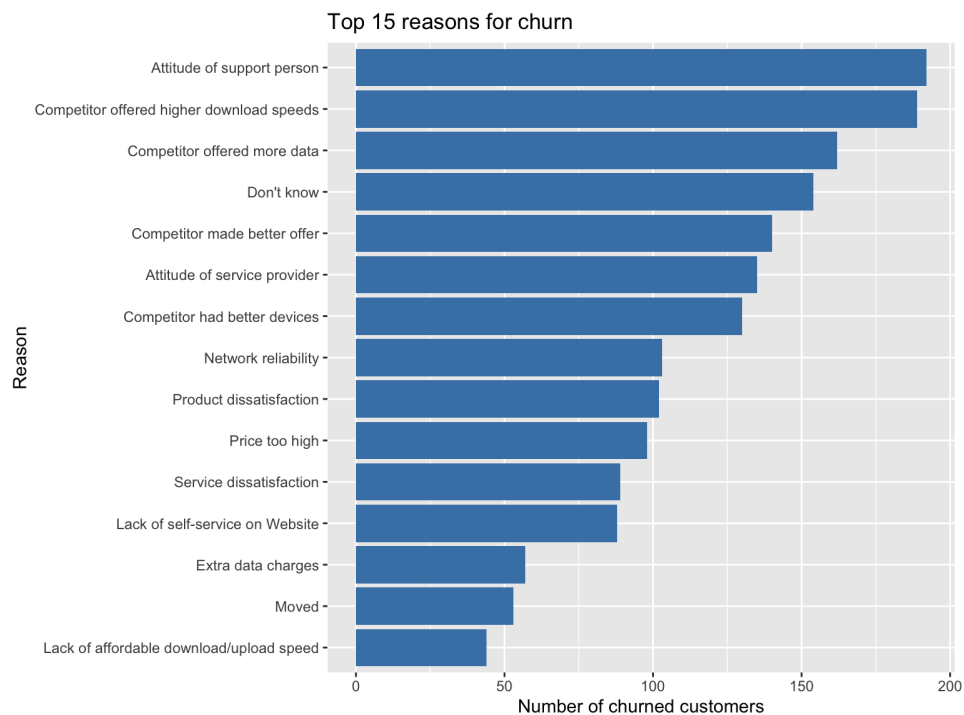
Q1. What reasons do churned customers give for leaving?

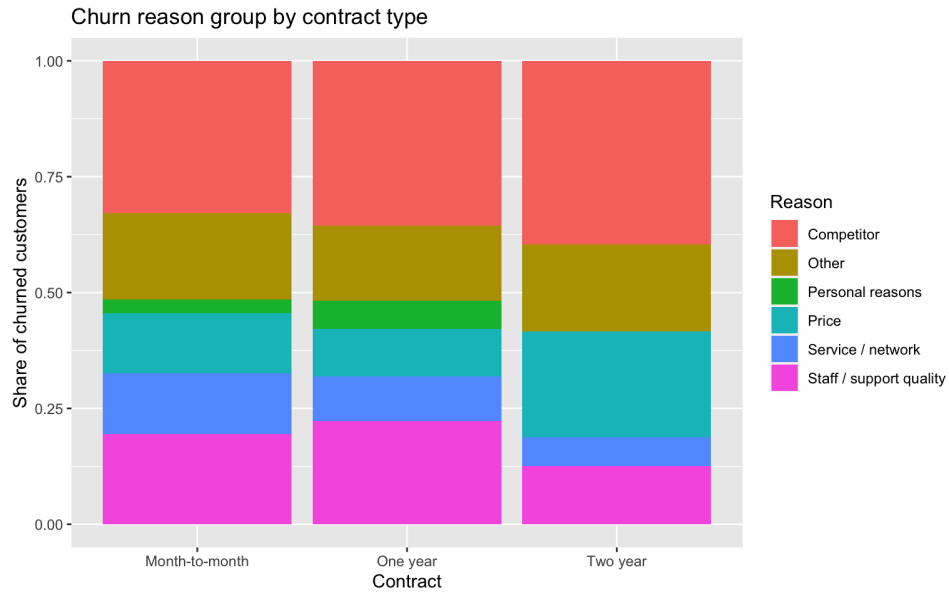
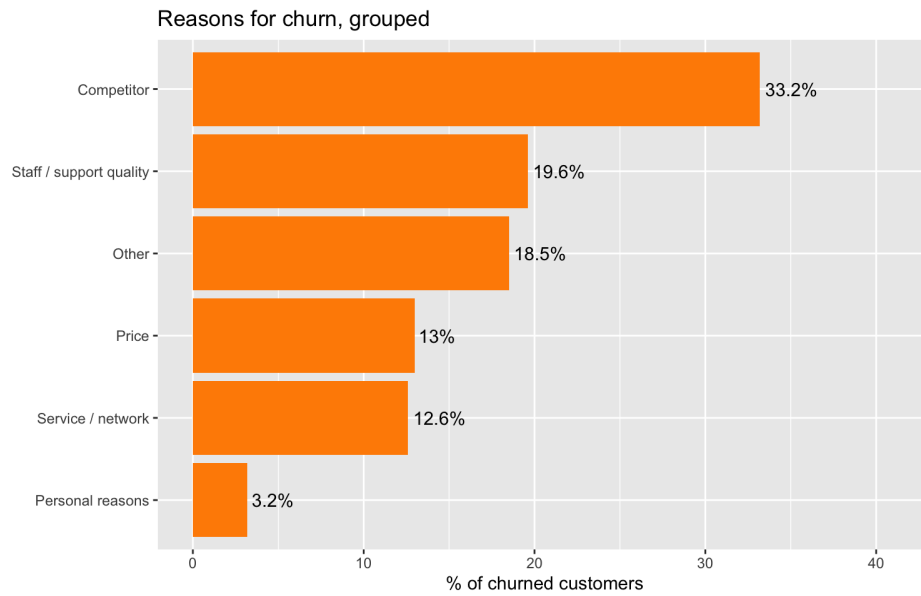
Answer: Mainly competitors (33.2%) and staff/support quality (19.6%). Personal reasons such as moving are only 3.2%.

Explanation

Of the 1,869 customers who churned, every one has a recorded reason (20 different ones). Grouping them shows that one third of churn is about a competitor offering better prices, devices, speeds or more data, and one fifth is about the attitude or lack of expertise of support staff. Price on its own is 13% and network or service issues 12.6%.

The reason mix is similar across contract types, but 89% of churners (1,655 customers) are on month-to-month contracts, so that is where these reasons play out. Both main reasons are controllable: better competitive offers, and better training and service quality.





Question 2

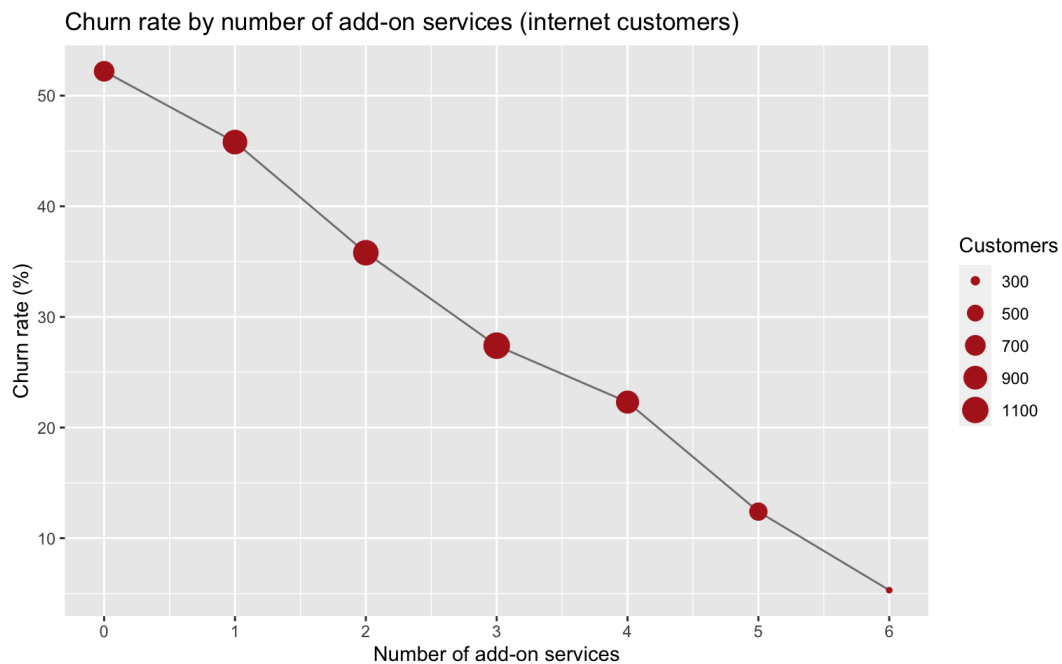
Q2. Do customers with more add-on services churn less?

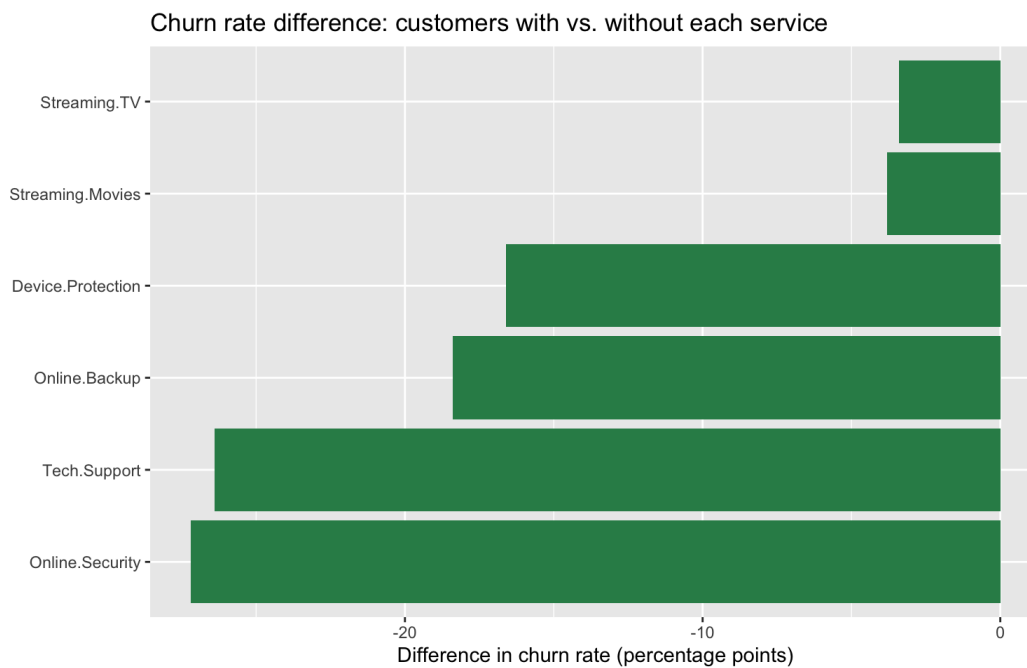
Answer: Yes. Churn falls steadily from 52% (no add-ons) to 5% (all six), but the type of service matters: Online Security and Tech Support are the most protective.

Explanation

Among internet customers, each extra service lowers churn (trend test $p < 2.2e-16$). Customers with 2 or more services churn much less than those with 0-1 (chi-square $p < 2.2e-16$). Looking at services one by one, Online Security (-27 points) and Tech Support (-26 points) show the biggest gaps, while the two streaming services show only -3 to -4 points.

Caution: this is an association. Customers who are already satisfied may simply buy more services.





Question 3

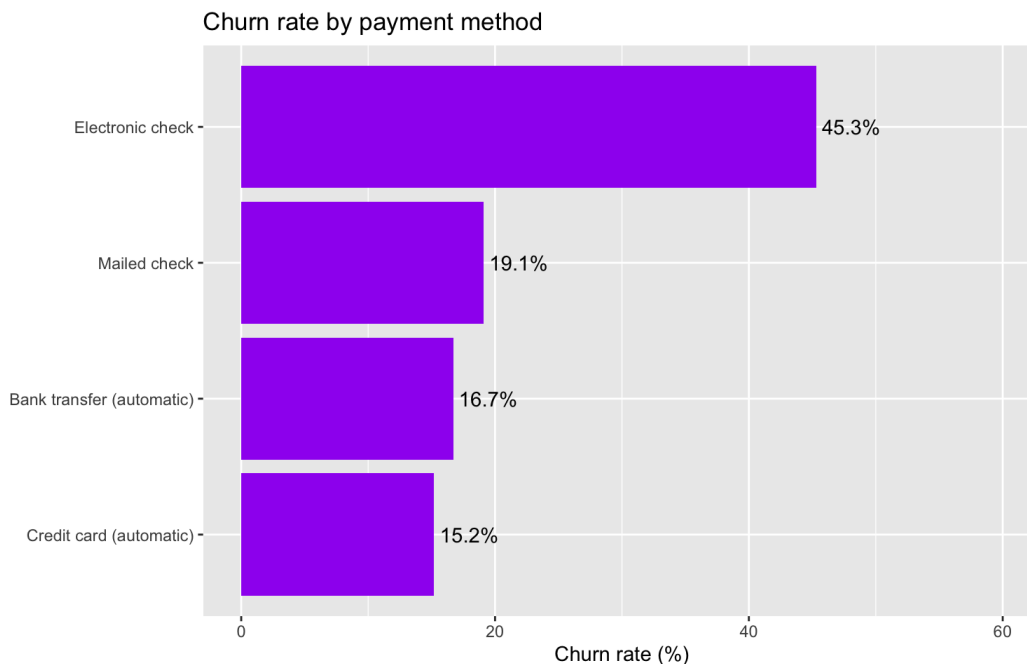
Q3. Are billing choices (payment method, paperless billing) linked to churn?

Answer: Yes. Electronic check users churn at 45.3% versus 15-19% for other methods, and the gap remains inside every contract type. Paperless billing is also linked to higher churn (33.6% vs. 16.3%).

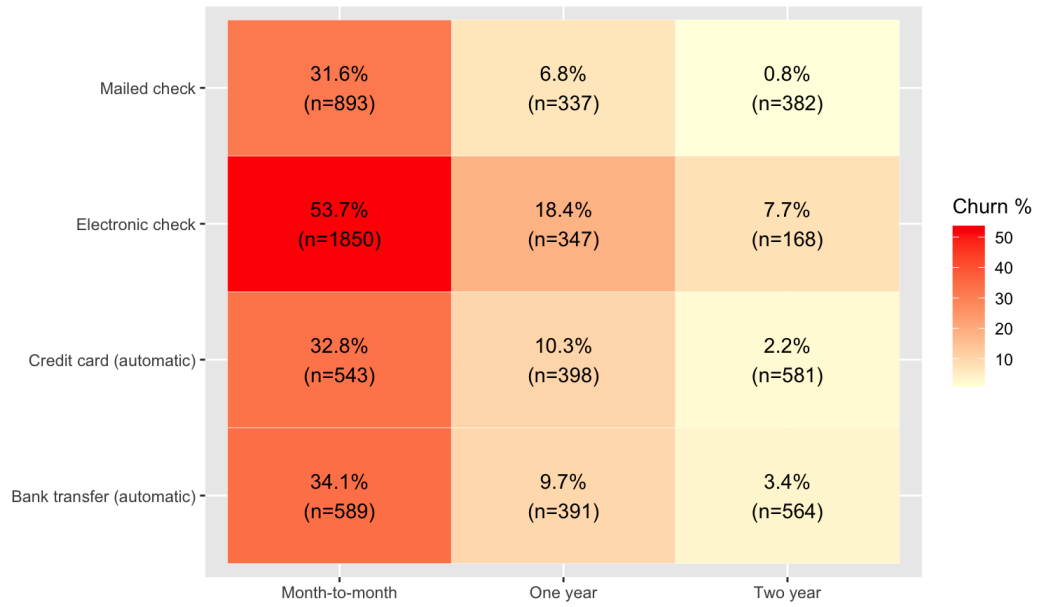
Explanation

Both chi-square tests are highly significant (payment $p < 2.2e-16$, paperless $p < 2.2e-16$). The heatmap shows that electronic check has the highest churn within month-to-month (53.7%), one-year (18.4%) and two-year (7.7%) contracts, so the effect is not only due to contract length.

Automatic payments (bank transfer, credit card) have the lowest churn. A possible reason is that automatic payment is a low-effort habit, so leaving takes an active decision.



Churn rate by payment method and contract



Question 4

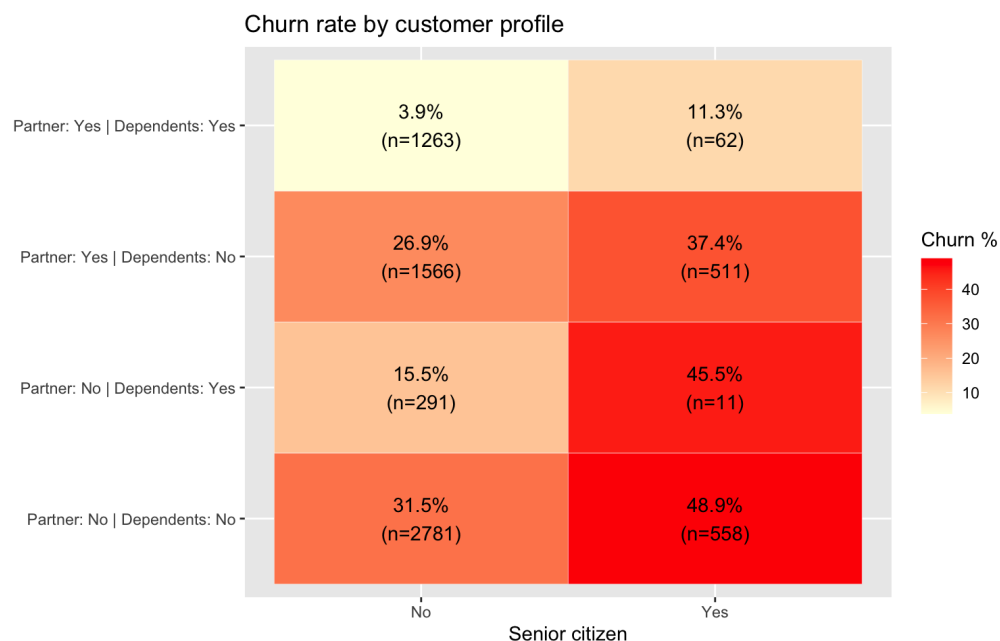
Q4. How do family situation and age relate to churn?

Answer: Seniors churn more (41.7% vs. 23.6%), customers with dependents churn far less (6.5% vs. 32.6%), and having a partner lowers churn (19.7% vs. 33.0%). Gender has no effect.

Explanation

The most at-risk profile is a senior with no partner and no dependents (48.9% churn, 558 customers). The safest is a non-senior with partner and dependents (3.9%, 1,263 customers). Chi-square tests are highly significant for senior, partner and dependents, and not significant for gender ($p = 0.49$).

In the combined model (Q8), dependents remain the strongest protective factor (odds ratio 0.24).



Question 5

Q5. Do churned customers pay more, and does it depend on the internet service type?

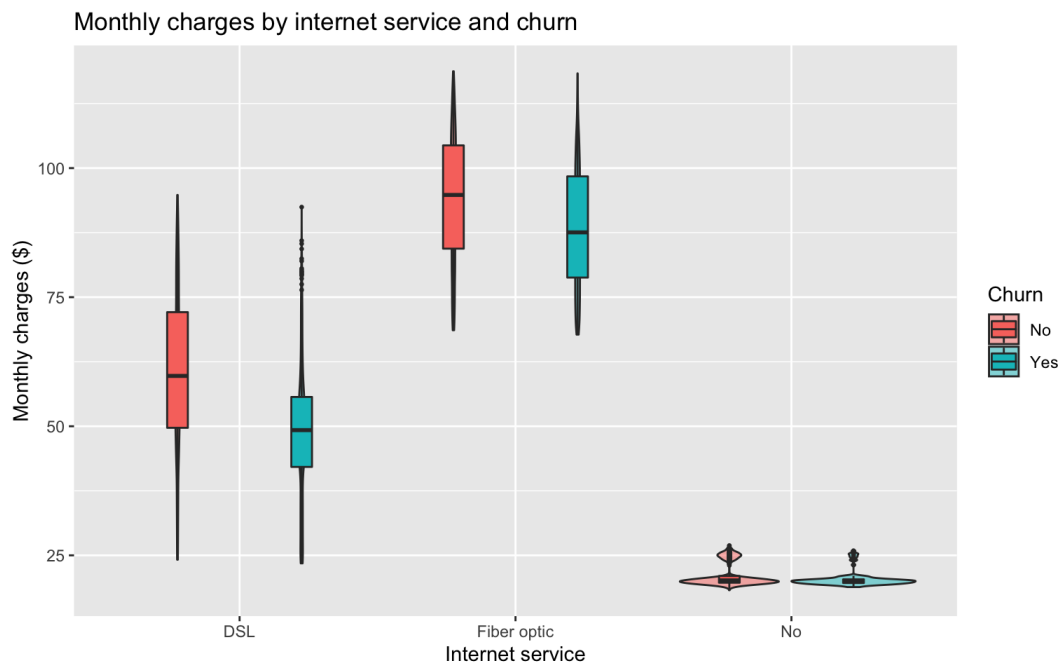
Answer: Overall yes (\$74 vs. \$61), but only because of the product mix. Inside each internet type, churners pay slightly less than those who stay.

Explanation

Fiber optic customers pay about \$90-95 per month and churn at 41.9%; DSL customers pay about \$55-60 and churn at 19.0%; customers without internet pay about \$20 and churn at 7.4%.

Because fiber customers are both expensive and likely to churn, the overall average for churners is higher. Within DSL, churners pay about \$11 less (mean \$49.1 vs. \$60.2); within fiber optic about \$6 less (\$88.1 vs. \$93.9). Wilcoxon tests are significant in all groups, but for customers without internet the gap is only \$0.70.

The conclusion is that the fiber optic product, not the price level itself, is associated with leaving. Competitor offers of higher download speed (Q1) may be part of the explanation.



Question 6

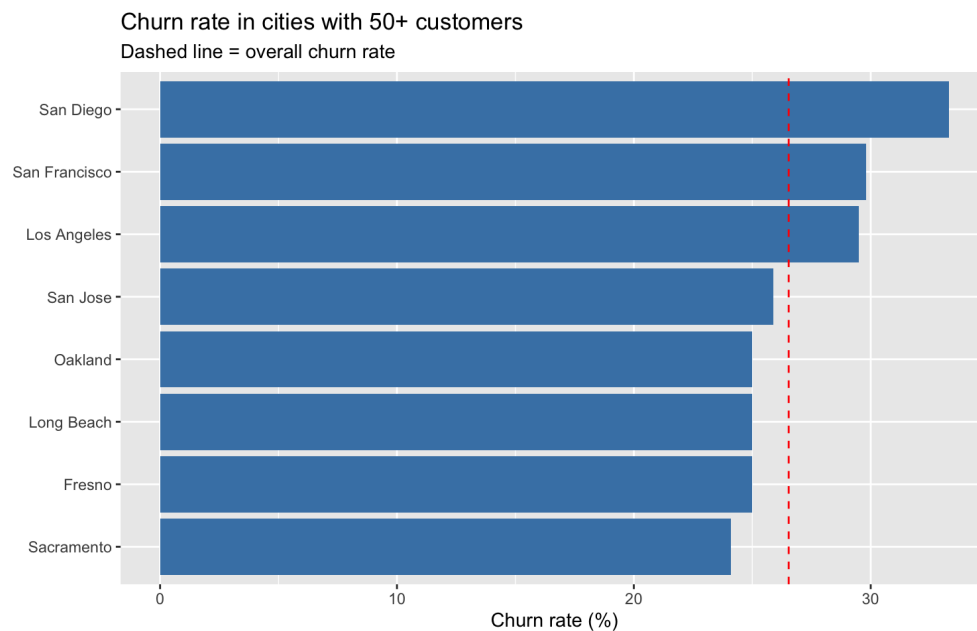
Q6. Does churn differ by location in California?

Answer: No meaningful difference. City churn rates range from 24% to 33%, but the differences are not statistically significant ($p = 0.73$).

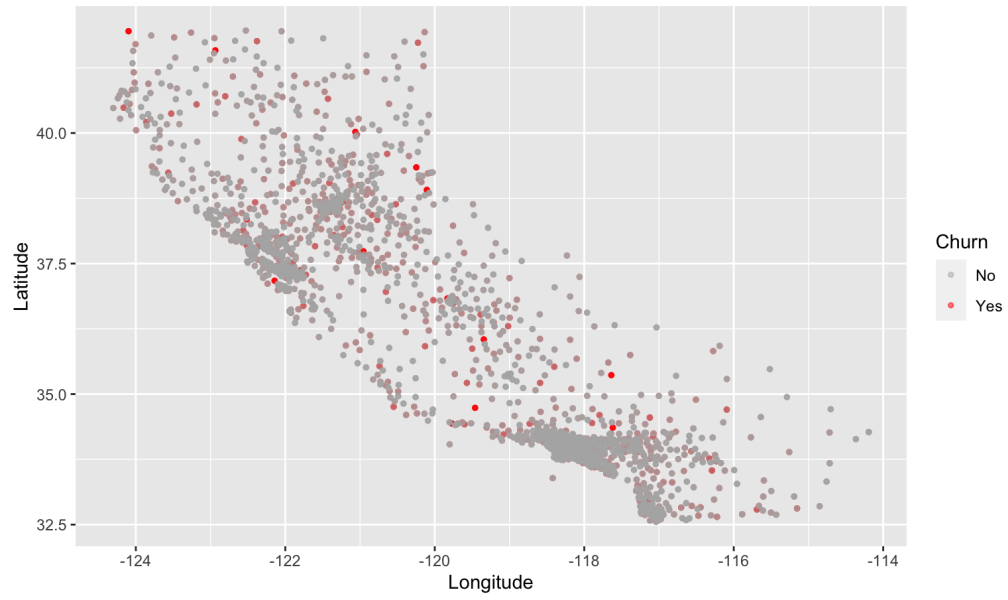
Explanation

We compared the eight cities with at least 50 customers. San Diego is highest at 33.3% and Sacramento lowest at 24.1%, around the overall 26.5%. The scatter map shows churned customers in every region of the state with no visible cluster.

Location is therefore not a main driver of churn compared with contract, services and billing.



Customers in California (churned in red)



Question 7

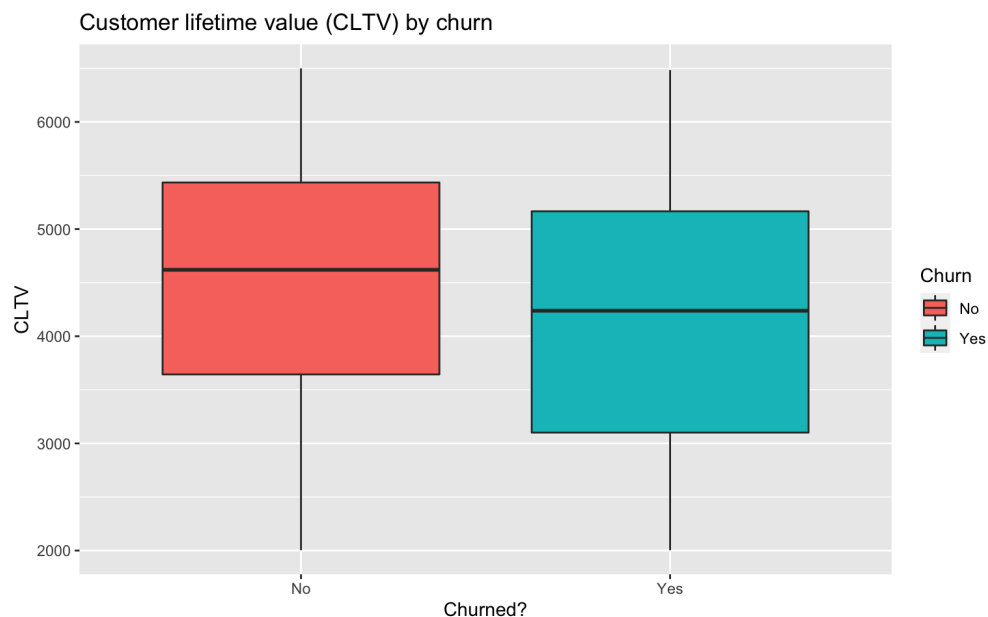
Q7. How much value is lost through churn?

Answer: Churned customers represent 30.5% of monthly revenue (\$139,131) while being 26.5% of customers, and 87% of the lost revenue is from month-to-month contracts.

Explanation

Churned customers pay more per month on average (\$74.4 vs. \$61.3) but have a lower lifetime value (mean CLTV \$4,149 vs. \$4,491, Wilcoxon $p < 2.2e-16$). The total CLTV of churned customers is about \$7.76 million. Of the lost monthly revenue, 86.9% comes from month-to-month, 10.1% from one-year and 3.0% from two-year contracts.

Note: IBM's *Churn Score* is strongly linked to the outcome (correlation 0.66). It was deliberately **not** used as a predictor, because it comes from IBM's own prediction and would cause data leakage.



Question 8

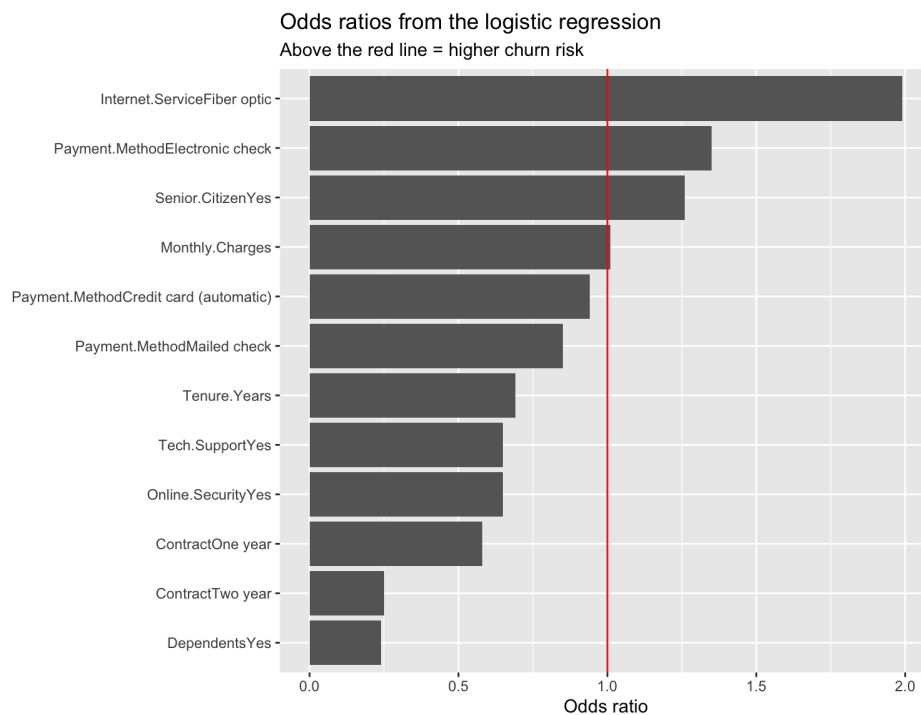
Q8. Which factors still matter when we look at them together?

Answer: Dependents, contract length, tenure, Online Security / Tech Support (all protective) and fiber optic internet, electronic check payment and senior age (all risky).

Explanation

A logistic regression on internet customers (70% training, 30% test, seed 123) gives odds ratios: fiber optic 1.99, electronic check 1.35, senior 1.26, tenure per year 0.69, Tech Support 0.65, Online Security 0.65, one-year contract 0.58, two-year contract 0.25 and dependents 0.24. Credit card and mailed check are not different from bank transfer, and monthly charges are only borderline significant ($p = 0.06$).

On the 30% of customers not used to fit the model, accuracy is about 77%, compared with 69% for always predicting 'No churn'. This is a useful check, not a final prediction model.



Summary of findings and implications

- **26.5%** of customers churned. Churn is mostly **preventable**: competitors (33%) and staff/support quality (20%) are the main reasons.
- **Protective services** (Online Security, Tech Support) go together with much lower churn (52% down to 5% as services increase).
- **Billing**: electronic check and paperless billing customers churn more, even within the same contract type.
- **Profile**: seniors living alone are highest risk; customers with dependents are safest. Gender does not matter.
- **Product, not price**: fiber optic customers churn at 41.9%. Within the same product, churners do not pay more.
- **Location** does not explain churn ($p = 0.73$).
- **Money**: churners are 30.5% of monthly revenue, 87% of it from month-to-month contracts.
- **Together**: dependents, contract length, tenure, fiber optic, electronic check and protective services are the key factors (77% test accuracy).

What the company could do

- Encourage month-to-month customers to move to longer contracts, especially in their first months.
- Promote Online Security and Tech Support, for example as a free trial for new customers.
- Review the fiber optic offer (price versus speed) and the quality of support staff.
- Offer incentives to switch from electronic check to automatic payment.
- Give special attention to senior customers who live alone.

Limitations and open questions

- The data is observational, so all results are associations, not proof of cause.
- Some groups are small (for example churners with long contracts and seniors with dependents but no partner).
- Churn reasons are self-reported; 154 customers answered 'Don't know'.
- Open questions: would early contact with tech support keep customers? Why does fiber optic churn more? Could a more flexible model (for example a decision tree) predict churn better?